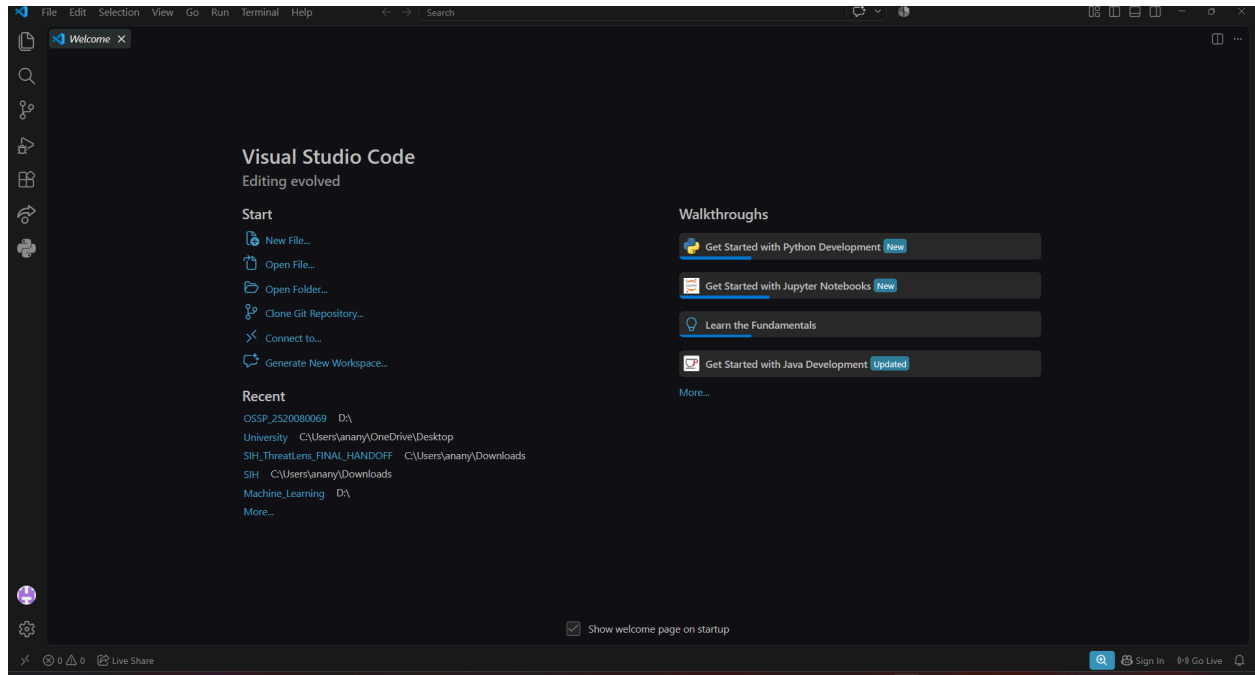
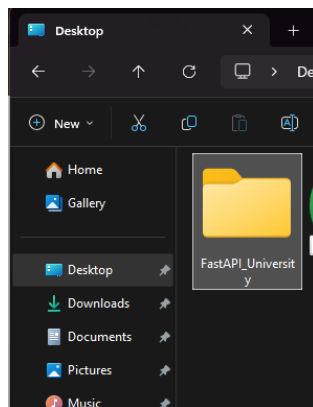


SKILL : WEEK - 5

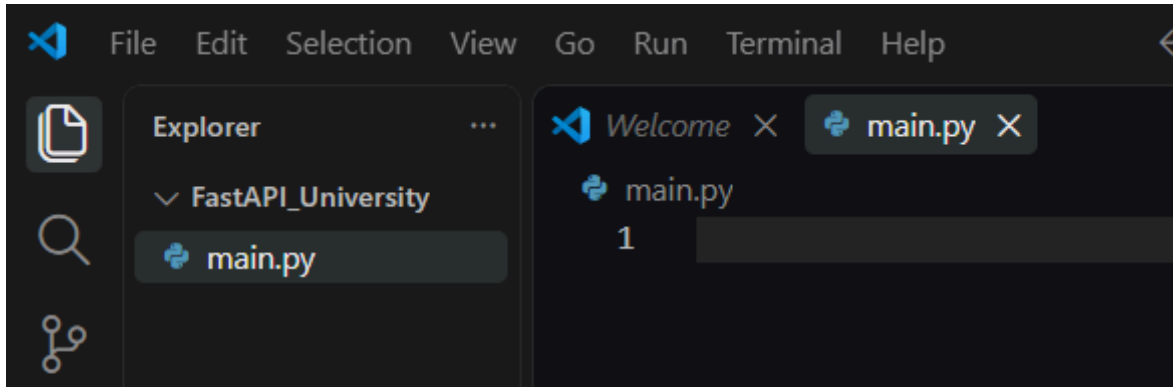
Step 1: Open VS Code



Step 2: Create a project folder

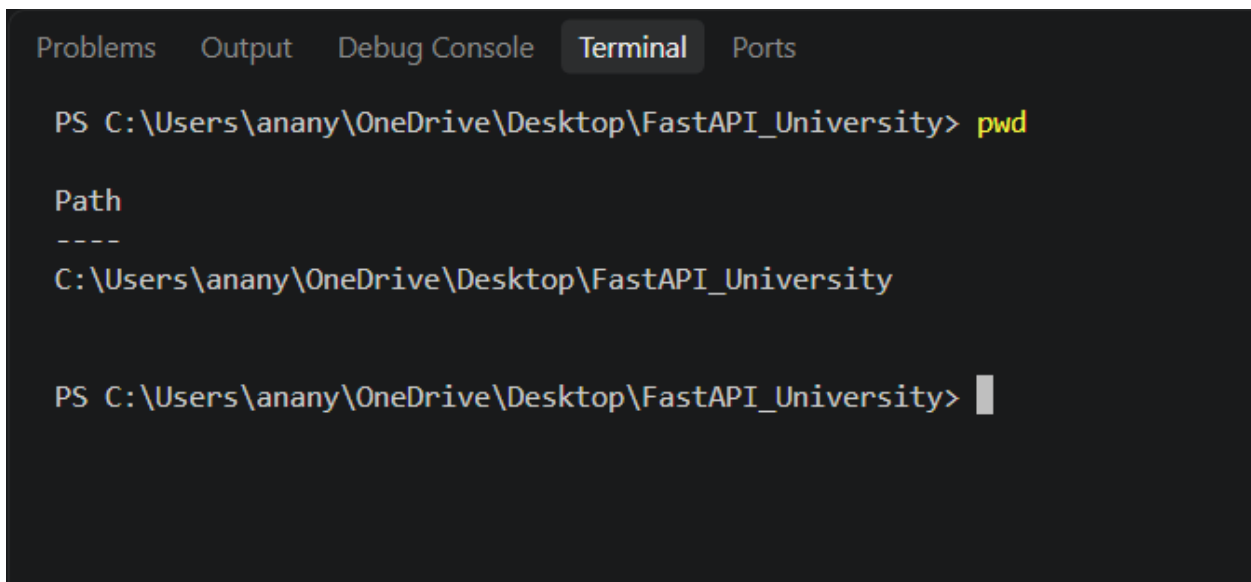


Step 3: Create [main.py](#)



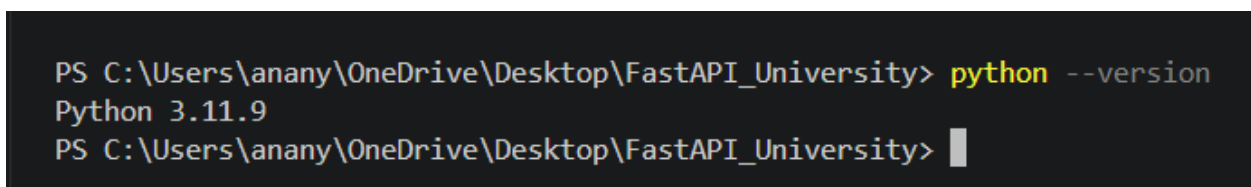
Step 4: Open the VS Code terminal

Pwd



Step 5: Check Python

python --version



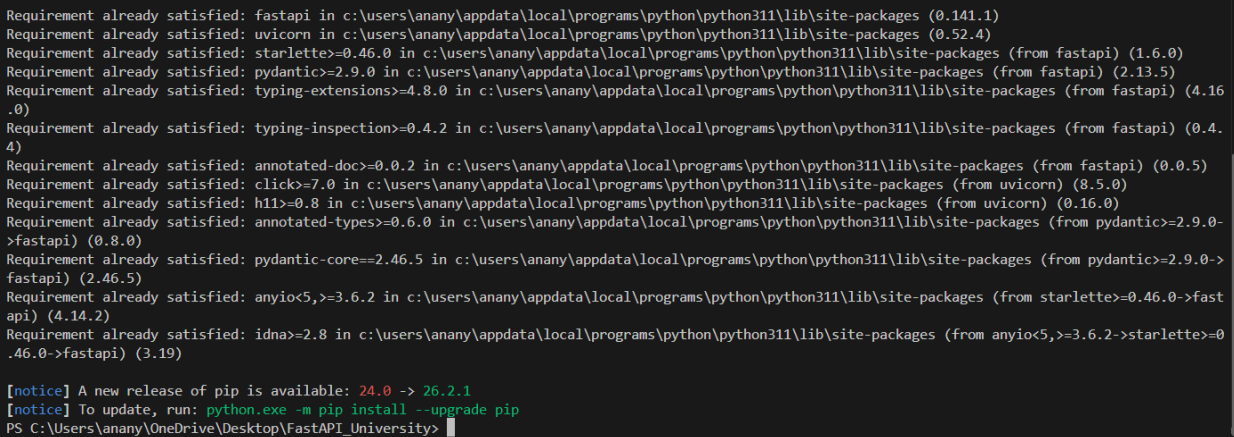
Step 6: Check pip

`python -m pip --version`

```
PS C:\Users\anany\OneDrive\Desktop\FastAPI_University> python -m pip --version
pip 24.0 from C:\Users\anany\AppData\Local\Programs\Python\Python311\Lib\site-packages\pip (python 3.11)
PS C:\Users\anany\OneDrive\Desktop\FastAPI_University>
```

Step 7: Install FastAPI and Uvicorn

`python -m pip install fastapi uvicorn`

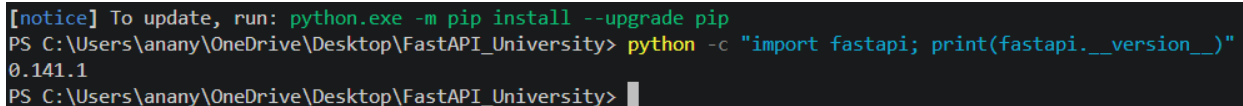


```
Problems Output Debug Console Terminal Ports
Requirement already satisfied: fastapi in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (0.141.1)
Requirement already satisfied: uvicorn in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (0.52.4)
Requirement already satisfied: starlette>=0.46.0 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from fastapi) (1.6.0)
Requirement already satisfied: pydantic>=2.9.0 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from fastapi) (2.13.5)
Requirement already satisfied: typing-extensions>=4.8.0 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from fastapi) (4.16.0)
Requirement already satisfied: typing-inspection>=0.4.2 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from fastapi) (0.4.4)
Requirement already satisfied: annotated-doc>=0.0.2 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from fastapi) (0.0.5)
Requirement already satisfied: click>=7.0 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from uvicorn) (8.5.0)
Requirement already satisfied: h11>=0.8 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from uvicorn) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from pydantic>=2.9.0->fastapi) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from pydantic>=2.9.0->fastapi) (2.46.5)
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from starlette>=0.46.0->fastapi) (4.14.2)
Requirement already satisfied: idna>=2.8 in c:\users\anany\appdata\local\programs\python\python311\lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi) (3.19)

[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\anany\OneDrive\Desktop\FastAPI_University>
```

Step 8: Verify FastAPI installation

`python -c "import fastapi; print(fastapi.__version__)"`



```
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\anany\OneDrive\Desktop\FastAPI_University> python -c "import fastapi; print(fastapi.__version__)"
0.141.1
PS C:\Users\anany\OneDrive\Desktop\FastAPI_University>
```

Step 9: Write your first FastAPI program

Open:

main.py

Write:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def home():
    return {
        "message": "Welcome to FastAPI"
    }
```

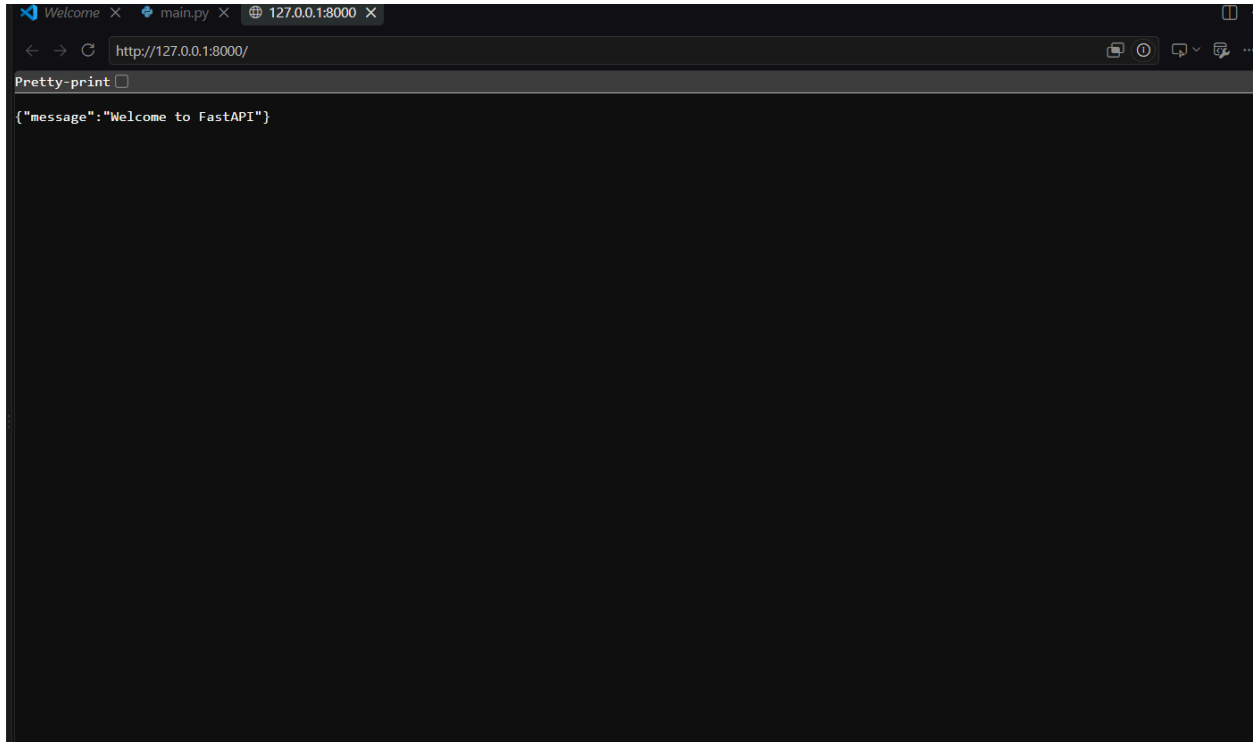
Step 10: Start the FastAPI server

python -m uvicorn main:app --reload



```
PS C:\Users\anany\OneDrive\Desktop\FastAPI_University> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\anany\\OneDrive\\Desktop\\FastAPI_University']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [31184] using WatchFiles
INFO: Started server process [14136]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Step 11: Open your API



Step 12: Open FastAPI documentation

FastAPI

0.1.0

OAS 3.1

/openapi.json

default

GET / Home

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to FastAPI" }</pre> Download Response headers <pre>content-length: 32 content-type: application/json date: Wed, 02 Sep 2026 09:30:17 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>"string"</pre>	No links

Step 13: Create your first Student API

```
# GET - Read students

@app.get("/students")

def get_students():

    return students
```

FastAPI 0.1.0 OAS 3.1

[/openapi.json](#)

default



GET

/students Get Students



Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/students' \
-H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/students

Server response

Code

Details

200

Response body

```
[
  {
    "id": 101,
    "name": "Ravi",
    "department": "CSE"
  },
  {
    "id": 102,
    "name": "Priya",
    "department": "CSE"
  }
]
```



Download

Response headers

```
content-length: 90
content-type: application/json
date: Wed,02 Sep 2026 09:32:46 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json



Controls Accept header.

Example Value | Schema

"string"

Step 14: Next learn POST

```
from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI()

@app.get("/students")
def get_students():
    return [
        {
            "id": 101,
            "name": "Ravi",
            "department": "CSE"
        },
        {
            "id": 102,
            "name": "Priya",
            "department": "CSE"
        }
    ]

class Student(BaseModel):
    id: int
    name: str
    department: str

@app.post("/students")
def create_student(student: Student):
    return {
        "message": "Student created",
        "student": student
    }
```

You can enter:

```
{
  "id": 103,
  "name": "Arun",
  "department": "ECE"
}
```

FastAPI

0.1.0

OAS 3.1

/openapi.json

default



GET /students Get Students



POST /students Create Student



Schemas



HTTPValidationError > Expand all [object](#)

Student > Expand all [object](#)

ValidationError > Expand all [object](#)

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "id": 103,
  "name": "Arun",
  "department": "ECE"
}'
```



Request URL

http://127.0.0.1:8000/students

Server response

Code

Details

200

Response body

```
{
  "message": "Student created",
  "student": {
    "id": 103,
    "name": "Arun",
    "department": "ECE"
  }
}
```



Download

Response headers

```
content-length: 83
content-type: application/json
date: Wed, 02 Sep 2026 09:36:14 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links
Media type application/json Controls Accept header.		
Example Value Schema		
<pre>"string"</pre>		
422	Validation Error	No links
Media type application/json		
Example Value Schema		
<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>		

Step 15: Then learn the complete CRUD

POST

```
@app.put("/students/{student_id}")
```

```
def update_student(student_id: int, student: Student):
```

```
    for i in range(len(students)):
```

```
        if students[i]["id"] == student_id:
```

```
            students[i] = student.dict()
```

```
            return {
```

```
                "message": "Student updated",
```

```
                "student": student
```

```
            }
```

```
    return {
```

```
        "message": "Student not found"
```

```
    }
```

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET

/students Get Students

POST

/students Create Student

PUT

/students/{student_id} Update Student

Schemas

HTTPValidationError > Expand all object

Student > Expand all object

ValidationError > Expand all object

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/students/102' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "id": 102,
  "name": "Priya",
  "department": "ECE"
}'
```

Request URL

http://127.0.0.1:8000/students/102

Server response

Code

Details

200

Response body

```
{
  "message": "Student updated",
  "student": {
    "id": 102,
    "name": "Priya",
    "department": "ECE"
  }
}
```



Download

Response headers

```
content-length: 84
content-type: application/json
date: Wed, 02 Sep 2026 09:47:30 GMT
server: uvicorn
```

Code	Description	Links
200	Successful Response	No links
Media type application/json Controls Accept header.		
Example Value Schema		
<pre>"string"</pre>		
422	Validation Error	No links
Media type application/json		
Example Value Schema		
<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>		

DELETE

```
@app.delete("/students/{student_id}")
def delete_student(student_id: int):

    for i in range(len(students)):

        if students[i]["id"] == student_id:

            deleted_student = students.pop(i)

            return {
                "message": "Student deleted",
                "student": deleted_student
            }

    return {
        "message": "Student not found"
    }
```

default

GET /students Get Students

POST /students Create Student

PUT /students/{student_id} Update Student

DELETE /students/{student_id} Delete Student

Schemas

HTTPValidationError > Expand all object

Curl

curl -X 'DELETE' \n'http://127.0.0.1:8000/students/102' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/students/102

Server response

Code

Details

200

Response body

{\n "message": "Student not found"\n}

Download

Response headers

content-length: 31\ncontent-type: application/json\ndate: Wed, 02 Sep 2026 09:50:04 GMT\nserver: uvicorn

Responses

Code

Description

Links

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"	No links
422	Validation Error Media type application/json Example Value Schema { "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }], }	No links